

Modern Physics for Secondary Schools
National Educational Curriculum Board
2026 Edition

Preface

This textbook introduces foundational physical concepts, mechanics, and gravitation for secondary students. Written in accordance with national pedagogical guidelines.

Acknowledgements

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Chapter 1. Motion and Kinematics

1.1 Introduction to Motion

Motion is the change in position of an object over time with respect to a reference frame.

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1.2 Speed and Velocity

Velocity is the directional speed of an object in motion as an indication of its rate of change in position.

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1.3 Acceleration and Free Fall

Acceleration is the rate of change of the velocity of an object with respect to time.

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1.4 Kinematics Exercises

Questions and problems on uniform acceleration in one dimension.

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Chapter 2. Laws of Force and Gravity

2.1 Newton's First Law

An object at rest remains at rest, and an object in motion remains in motion unless acted upon by a net external force.

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2.2 Newton's Second Law and Force

Force equals mass times acceleration ($F = ma$).

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2.3 Gravitational Attraction

Every particle attracts every other particle in the universe with a force proportional to the product of their masses.

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2.4 Gravity Exercises

Problems on planetary motion and circular orbits.

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Chapter 3. Work, Energy and Power

3.1 Concept of Work

Work is done when a force that is applied to an object results in a displacement in the direction of the force.

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3.2 Kinetic and Potential Energy

Kinetic energy is energy due to motion. Potential energy is energy stored in a system.

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3.3 Conservation of Energy

Energy can neither be created nor destroyed, only transformed from one form to another.

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3.4 Work and Energy Exercises

Practice questions and numerical calculations.

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Appendix: Answers to Selected Problems

Chapter 1 Solutions:

1. 15 m/s

2. 9.8 m/s^2

3. 45 meters

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Appendix Continued: Answers to Selected Problems

Chapter 2 Solutions:

1. 50 N
2. 120 kg m/s
3. 980 Joules

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